

A parser tells the difference between a variable declaration and a function declaration by checking what comes after the identifier. If the parser sees an opening parenthesis (right after the name, it recognizes it as a function declaration because parameters always appear in parentheses. If it sees an equals sign = or a semicolon ;, it treats it as a variable declaration instead. For example:

```
int add(int a, int b) {  
    return a + b;  
}
```

Here, the parser reads int as a type, add as an identifier, and when it finds the (, it knows this must be a function. It then parses the parameters (int a, int b) and the body { return a + b; }. If it instead found int a = 5;, the lack of parentheses after a would cause the parser to handle it as a variable declaration.

https://www.reddit.com/r/ProgrammingLanguages/comments/u9mm3x/lets_talk_function_declarations/